

GENETIC ENGINEERING: GM CROPS AND CRISPR

BREEDING AND ENGINEERING TAXONOMY → BT COTTON MECHANISM AND → DMH-11 CONSTRUCT AND STATUS → CRISPR TARGETING RAIL → EDITING-OUTCOME MATRIX → SDN CLASSIFICATION LADDER

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g3 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 BREEDING AND ENGINEERING TAXONOMY

BREEDING AND ENGINEERING TAXONOMY

CONVENTIONAL CROSS

PARENTAL GENOMES RECOMBINE SEXUALLY

OFFSPRING OF SELECTED PARENTAL LINES

SEXUALLY INCOMPATIBLE DONOR DNA

INTACT COMPATIBLE-POOL GENE

REARRANGED COMPATIBLE-POOL ELEMENTS

CONVENTIONAL CROSS → parental genomes recombine sexually

HYBRID → offspring of selected parental lines

TRANSGENIC → sexually incompatible donor DNA

CISGENIC → intact compatible-pool gene

CONCEPT / ARCHITECTURE

- CONVENTIONAL CROSS → parental genomes recombine sexually
- HYBRID → offspring of selected parental lines

MECHANISM / APPLICATION

- TRANSGENIC → sexually incompatible donor DNA
- CISGENIC → intact compatible-pool gene

READINESS / STATUS LIMIT

- INTRAGENIC → rearranged compatible-pool elements
- CONVENTIONAL CROSS → parental genomes recombine sexually

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CISGENIC → intact compatible-pool gene; INTRAGENIC → rearranged compatible-pool elements.

01

STAGE 01 BT COTTON MECHANISM AND STEWARDSHIP

BT COTTON MECHANISM AND STEWARDSHIP

CRY GENE

PLANT EXPRESSES BT PROTEIN

SPECIFIC BOLLWORM FEEDING

TOXIC ACTION IN SUSCEPTIBLE LARVAE

SELECTION PRESSURE

RESISTANT ALLELES CAN INCREASE

REFUGE + STACKING

CRY GENE → plant expresses Bt protein

SPECIFIC BOLLWORM FEEDING → toxic action in susceptible larvae

SELECTION PRESSURE → resistant alleles can increase

REFUGE + STACKING → slow resistance evolution

TRAP → insect resistance is not herbicide tolerance

STARTING CONDITION / INPUT

- CRY GENE → plant expresses Bt protein
- SPECIFIC BOLLWORM FEEDING → toxic action in susceptible larvae

SCIENTIFIC OR ENGINEERING MECHANISM

- SELECTION PRESSURE → resistant alleles can increase
- REFUGE + STACKING → slow resistance evolution

OUTPUT / FAILURE BOUNDARY

- TRAP → insect resistance is not herbicide tolerance
- CRY GENE → plant expresses Bt protein

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → insect resistance is not herbicide tolerance.

02

STAGE 02 DMH-11 CONSTRUCT AND STATUS

DMH-11 CONSTRUCT AND STATUS

MALE STERILITY IN ONE PARENT

FERTILITY RESTORATION IN HYBRID

SELECTABLE MARKER

ENABLE HYBRIDISATION IN MUSTARD

NO COMMERCIAL-CULTIVATION OR SETTLED-LEGALITY CLAIM

BARNASE → male sterility in one parent

BARSTAR → fertility restoration in hybrid

BAR → selectable marker

PURPOSE → enable hybridisation in mustard

CONCEPT / ARCHITECTURE

- BARNASE → male sterility in one parent
- BARSTAR → fertility restoration in hybrid

MECHANISM / APPLICATION

- BAR → selectable marker
- PURPOSE → enable hybridisation in mustard

READINESS / STATUS LIMIT

- STATUS → no commercial-cultivation or settled-legality claim
- BARNASE → male sterility in one parent

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PURPOSE → enable hybridisation in mustard; STATUS → no commercial-cultivation or settled-legality claim.

03

STAGE 03 CRISPR TARGETING RAIL

CRISPR TARGETING RAIL

GUIDE RNA

SEQUENCE RECOGNITION

TARGET DNA + PAM

PERMITTED BINDING CONTEXT

DOUBLE-STRAND CLEAVAGE

CELLULAR REPAIR

CREATES FINAL EDIT

GUIDE RNA → sequence recognition

TARGET DNA + PAM → permitted binding context

CAS9 → double-strand cleavage

CELLULAR REPAIR → creates final edit

RULE → targeting tool and repair outcome are distinct

STARTING CONDITION / INPUT

SCIENTIFIC OR ENGINEERING MECHANISM

OUTPUT / FAILURE BOUNDARY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PURPOSE → enable hybridisation in mustard; STATUS → no commercial-cultivation or settled-legality claim.

03

STAGE 03 CRISPR TARGETING RAIL

CRISPR TARGETING RAIL GUIDE RNA SEQUENCE RECOGNITION TARGET DNA + PAM PERMITTED BINDING CONTEXT DOUBLE-STRAND CLEAVAGE CELLULAR REPAIR CREATES FINAL EDIT

GUIDE RNA → sequence recognition

TARGET DNA + PAM → permitted binding context

CAS9 → double-strand cleavage

CELLULAR REPAIR → creates final edit

RULE → targeting tool and repair outcome are distinct

STARTING CONDITION / INPUT

- GUIDE RNA → sequence recognition
- TARGET DNA + PAM → permitted binding context

SCIENTIFIC OR ENGINEERING MECHANISM

- CAS9 → double-strand cleavage
- CELLULAR REPAIR → creates final edit

OUTPUT / FAILURE BOUNDARY

- RULE → targeting tool and repair outcome are distinct
- GUIDE RNA → sequence recognition

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → targeting tool and repair outcome are distinct.

04

STAGE 04 EDITING-OUTCOME MATRIX

EDITING-OUTCOME MATRIX TEMPLATE-FREE INDELS AND KNOCKOUT DONOR-TEMPLATE SUBSTITUTION OR INSERTION BASE EDITING CHEMICAL BASE CONVERSION WITHOUT DSB PRIME EDITING TEMPLATED SHORT WRITING WITHOUT DSB PRECISION MECHANISMS ARE NOT INTERCHANGEABLE

SYSTEM / DEVICE / INSTITUTION

NHEJ → template-free indels and knockout
HDR → donor-template substitution or insertion

DECISIVE TECHNICAL DISTINCTION

BASE EDITING → chemical base conversion without DSB
PRIME EDITING → templated short writing without DSB

STATUS / UNIT / EVIDENCE LIMIT

TRAP → precision mechanisms are not interchangeable
NHEJ → template-free indels and knockout

SYSTEM / DEVICE / INSTITUTION

- NHEJ → template-free indels and knockout
- HDR → donor-template substitution or insertion

DECISIVE TECHNICAL DISTINCTION

- BASE EDITING → chemical base conversion without DSB
- PRIME EDITING → templated short writing without DSB

STATUS / UNIT / EVIDENCE LIMIT

- TRAP → precision mechanisms are not interchangeable
- NHEJ → template-free indels and knockout

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PRIME EDITING → templated short writing without DSB.

05

STAGE 05 SDN CLASSIFICATION LADDER

SDN CLASSIFICATION LADDER NO TEMPLATE; SMALL EDIT SHORT TEMPLATE; DEFINED SMALL EDIT GENE-SIZED TARGETED INSERTION 2022 LIGHTER TRACK SDN-1/2 FREE OF EXOGENOUS DNA FULL SCRUTINY SDN-3 REMAINS OUTSIDE EXEMPTION

SDN-1 → no template; small edit

SDN-2 → short template; defined small edit

SDN-3 → gene-sized targeted insertion

2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA

STARTING CONDITION / INPUT

- SDN-1 → no template; small edit
- SDN-2 → short template; defined small edit

SCIENTIFIC OR ENGINEERING MECHANISM

- SDN-3 → gene-sized targeted insertion
- 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA

OUTPUT / FAILURE BOUNDARY

- FULL SCRUTINY → SDN-3 remains outside exemption
- SDN-1 → no template; small edit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 2022 LIGHTER TRACK → SDN-1/2 free of exogenous DNA.

06

STAGE 06 SIX-TIER BIOSAFETY INSTITUTION MAP

SIX-TIER BIOSAFETY INSTITUTION MAP RECOMBINANT-DNA ADVICE INSTITUTIONAL DAY-TO-DAY OVERSIGHT RESEARCH AND CONTAINED STAGES ENVIRONMENTAL RELEASE AND LARGE-SCALE USE SBCC + DLC STATE AND DISTRICT MONITORING

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversight

RCGM / DBT → research and contained stages

GEAC / MoEF&CC → environmental release and large-scale use

CONCEPT / ARCHITECTURE

- RDAC → recombinant-DNA advice
- IBSC → institutional day-to-day oversight

MECHANISM / APPLICATION

- RCGM / DBT → research and contained stages
- GEAC / MoEF&CC → environmental release and large-scale use

READINESS / STATUS LIMIT

- SBCC + DLC → state and district monitoring
- RDAC → recombinant-DNA advice

06

STAGE 06

SIX-TIER BIOSAFETY INSTITUTION MAP

SIX-TIER BIOSAFETY INSTITUTION MAPRECOMBINANT-DNA ADVICERDACCORE GMO BIOSAFETY FRAMEWORKCONTAINED WORKENVIRONMENTAL APPRAISALGM-FOOD SAFETY APPROVAL AND LABELLING TRACKCROP SYSTEMAGRONOMIC EVALUATION AND VARIETY PROCESSES

RDAC → recombinant-DNA advice

IBSC → institutional day-to-day oversightRCGM / DBT → research and contained stagesGEAC / MoEF&CC → environmental release and large-scale use

CONCEPT / ARCHITECTUREMECHANISM / APPLICATIONREADINESS / STATUS LIMIT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SBCC + DLC → state and district monitoring.

07

STAGE 07

LAW AND REGULATOR SPLIT

LAW AND REGULATOR SPLITRULES 1989CORE GMO BIOSAFETY FRAMEWORKCONTAINED WORKENVIRONMENTAL APPRAISALGM-FOOD SAFETY APPROVAL AND LABELLING TRACKCROP SYSTEMAGRONOMIC EVALUATION AND VARIETY PROCESSES

SYSTEM / DEVICE / INSTITUTIONDECISIVE TECHNICAL DISTINCTIONSTATUS / UNIT / EVIDENCE LIMIT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ICAR / CROP SYSTEM → agronomic evaluation and variety processes.

08

STAGE 08

INDIAN DECISION TIMELINE

INDIAN DECISION TIMELINE09 FEB 2010BT BRINJAL MORATORIUM30 MAR 2022SDN-1/2 EXEMPTION OM17 MAY 2022DBT GENOME-EDITED PLANT GUIDELINES23 JUL 2024

09 FEB 2010 → Bt brinjal moratorium30 MAR 2022 → SDN-1/2 exemption OM17 MAY 2022 → DBT genome-edited plant guidelines23 JUL 2024 → Supreme Court DMH-11 split verdict04 APR 2025 → two genome-edited rice varieties announced

CONCEPT / ARCHITECTUREMECHANISM / APPLICATIONREADINESS / STATUS LIMIT

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: 23 JUL 2024 → Supreme Court DMH-11 split verdict.

09

STAGE 09

BIOSAFETY MECHANISM MAP

BIOSAFETY MECHANISM MAPTARGET PEST PRESSURERESISTANCE EVOLUTIONSEED MOVEMENTGENE-FLOW EXPOSURETRAIT EXPRESSIONNON-TARGET EXPOSURE PATHWAYSIMILAR DNA SITES

TARGET PEST PRESSURE → resistance evolution

POLLEN / SEED MOVEMENT → gene-flow exposureTRAIT EXPRESSION → non-target exposure pathwaySIMILAR DNA SITES → possible off-target edit

STARTING CONDITION / INPUTSCIENTIFIC OR ENGINEERING MECHANISMOUTPUT / FAILURE BOUNDARY

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Genetic engineering changes DNA through defined tools; GM crops and CRISPR.

10

STAGE 10

FARMER-LIVELIHOOD EVALUATION

